

# Samar Kansal

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## SUMMARY

Software engineer with 4+ years designing and shipping backend systems that handle millions of daily requests across distributed, production environments. Enjoys the hard problems in scale, reliability, and real-time data, and cares as much about the team understanding the system as building it right.

## EXPERIENCE

### Software Development Engineer | Amazon | Washington, D.C | May 2025 - Present

- Developed Java microservices and **real-time pub/sub** infrastructure for **AWS AppSync**, a managed GraphQL/Event API service handling millions of concurrent Websocket connections across **34 AWS regions** - building event broadcast pipelines on EC2, SQS, DynamoDB, and NLB/ALB within a fault-isolated multi-cell architecture with canary testing.
- Designed and built a **serverless Execution Service** (Java, API Gateway, Lambda, DynamoDB, SQS, EventBridge) from scratch, processing 500+ daily pipeline/ML model orchestration requests with input validation, cross-account event-driven notifications, and semantic version resolution, serving 12 teams with p99 latency under 3s at 99.9% uptime.
- Engineered core components of CMOF**, a centralized ML platform supporting 20+ models, reducing production time by 60% through automated IaC and multi-environment CI/CD pipelines that slashed deployment cycles from 2 weeks to 2 days while maintaining strict governance.
- Architected **high-throughput data pipeline** using AWS Step Functions and Glue PySpark to orchestrate the processing of **datasets scaled to 800B+ records**, migrating legacy Redshift-based queries into a centralized SageMaker Feature Store. Added automated validation rules and CloudWatch alerts, reducing manual setup from days to hours.
- Designed a **Python-based Model Development Kit (SDK)** that automates Sagemaker image builds and remote execution by handling dependency resolution and S3 artifact packaging, allowing users to hot-deploy local code from Jupyter notebooks directly to AWS for rapid, iterative prototyping.

### Software Engineer 1-2 | ServiceNow | Hyderabad, India | August 2020 - December 2022

- Architected **REST and GraphQL APIs** powering the Service Operations Workspace for ITSM/ITOM use cases (e.g., incident, alert, and problem management) serving **5M+ users**; designed scalable data models enabling **2M+ daily transactions** with sub-millisecond **MySQL** access latency.
- Developed a **multi-channel communication platform** (Java, REST, WebSockets) for **on-call and incident management workflows**, integrating **Twilio, Zoom, and Microsoft Teams** to automate **10K+ daily alerts** across SMS, email, and voice channels with **99.9% uptime**.
- Developed a DevOps/SRE management application on the **ServiceNow platform (JavaScript compiled to Java on the Rhino engine)** using **React, GraphQL, and MySQL**; implemented backend services integrating AWS and Azure to automate alerts and create incidents/changes based on user-defined rules.

### Software Developer (GA) | Virginia Tech: Division of IT | January 2023 - April 2025

- Built and deployed a Spring Boot + React equipment-loaning platform for 50K+ students, containerized with Docker and orchestrated on Kubernetes.
- Migrated 500K+ users from Gmail to Outlook, building secure GraphQL + Angular components integrated with Microsoft Azure APIs via OAuth 2.0.

### Software Developer Intern | ServiceNow | Hyderabad, India | July 2019 - December 2019

- Enhanced ITSM apps for **15K+ enterprises** using **JavaScript, GlideRecord, jQuery**, and **GlideAjax**, improving query performance by **25%**.
- Built a React calendar component adopted by 30+ teams; achieved 85% test coverage via Selenium + Jenkins CI/CD.

### Software Developer Intern | AlphaICs | Bangalore, India | May 2019 - June 2019

- Created custom ML operations (OPs) for TensorFlow with C++ achieving 30% faster inference on edge devices.
- Implemented parallel programming techniques, reducing computation by 3x for Molecular Dynamics using OpenMP and CUDA.

## SKILLS

**Languages:** Java, Python, JavaScript/TypeScript, SQL, C/C++, Kotlin

**Backend & APIs:** Spring Boot, Node.js, REST, GraphQL, gRPC, WebSockets, Microservices, Serverless, FastAPI, Event-driven Architecture

**AWS:** EC2, Lambda, API Gateway, SQS, SNS, EventBridge, Step Functions, Glue, SageMaker, Kinesis, DynamoDB, Aurora, S3, CloudWatch

**Data & Messaging:** Kafka, PySpark, Databricks, Redis, MySQL, PostgreSQL, MongoDB, Redis, Redshift

**DevOps:** Docker, Kubernetes, CI/CD, IaC, Terraform, Canary Deployments, Git

**AI/ML:** SageMaker Feature Store, MLOps, Model Orchestration, LLM Integration, Autogen

## CERTIFICATIONS

**AWS Solutions Architect, AWS Developer Associate**

## EDUCATION

**Masters in Computer Science | Virginia Tech | Blacksburg, Virginia | 2024 | 4.00**

**Bachelors in Electronics and Communication | BITS Pilani | Hyderabad, India | 2020 | 3.55**

## PROJECTS

### Attendify: Meeting scheduler and QR-based attendance management system | | [youtu.be/KfypQMR2\\_yM](https://youtu.be/KfypQMR2_yM)

- Built a meeting scheduler, integrated with Google Calendar to send invitations to 300 members with unique, time-sensitive QR codes.
- Members can use the QR code to mark attendance securely, and organizers can generate tamper-proof attendance reports.
- Technology:** Vue.js, MongoDB, Express, Pinia for state management, JWT for authentication, containerized with Docker and Kubernetes.

### CarCool: ML-Powered Ride Sharing Platform | | [mycarcool.netlify.app/](https://mycarcool.netlify.app/)

- Developed a smart ride-sharing app with user-profile customization for lifestyle-based compatibility matching, similar to dating platforms.
- Enabled drivers to create location-optimized rides powered by Google Maps and passengers to find nearby carpool options with "Vibe Scores" display.
- Technology:** Python, FastAPI for RESTful APIs, MongoDB with geospatial indexing and geohashing, AWS EC2 for deployment with auto-scaling, Vue.js.